The Effects of a Linear Dural Incision on 60-Day Postoperative Complications for Chiari I Malformation Decompression

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BACKGROUND

Chiari malformation type I (CM-I) is defined by caudal descent of the cerebellar tonsils at least 5 mm below the foramen magnum, which obstructs cerebrospinal fluid (CSF) flow at the craniocervical junction.[1] Posterior fossa decompression (PFD) restores flow; the standard approach pairs suboccipital craniectomy and C1 laminectomy with a Y-shaped dural incision and expansile duraplasty.[2] Despite symptomatic improvement in most patients, postoperative complications occur in 11–34% of patients.[3,4] CSF leak, pseudomeningocele, and aseptic meningitis predominate,[5,6] and each prolongs hospitalization,[7] increases reoperation,[8] and blunts long-term gains in pain, disability, and quality of life.[9]

Dural incision geometry offers a mechanistic target. The Y-incision produces a long, multidirectional suture line with three high-tension apices that are likely susceptible to dehiscence under pulsatile CSF pressure. Prior efforts to reduce postoperative complications have targeted closure technique and graft type, but the dural opening itself has received little attention.

Purpose:

To compare 60-day postoperative composite complications between Y-shaped and vertical linear dural incisions in adults undergoing PFD for CM-I.

Hypothesis:

We hypothesized that the linear incision would be associated with a lower 60-day composite complication rate because a single closure line eliminates the high-tension apices of the Y-incision.

METHODS

Participants:

- Retrospective cohort of 36 consecutive adults with symptomatic CM-I who underwent PFD (31 F, 5 M; age 42.4 ± 11.8 yr).
- Inclusion: age ≥ 18 yr, tonsillar descent ≥ 5 mm, symptomatic CM-I.

Surgical Technique:

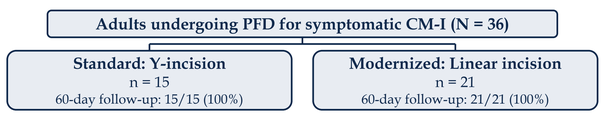
- Patients were placed prone, secured in a Mayfield® skull clamp (Integra LifeSciences, Princeton, NJ), and prepared in standard sterile fashion.
- All patients underwent suboccipital craniectomy with C1 laminectomy.
- Dural incision shape defined group assignment.
- Subarachnoid dissection and tonsillar cautery were performed in all linear-incision cases and in most Y-incision cases (86.7% and 73.3%; Table 1); watertight closure with or without duraplasty was standard in both groups.
- Duraplasty was performed at the surgeon's discretion; a dural onlay over a watertight primary closure was not counted as duraplasty.
- Standard muscle, fascia, and skin closure was performed in all patients.

Outcomes:

- **Primary:** 60-day composite complication, defined as any of the following within 60 days: CSF leak, pseudomeningocele, meningitis, surgical site infection, hemorrhage, hydrocephalus, new neurological deficit, elevated ICP/herniation, DVT/PE, or systemic infection.
- **Secondary:** duraplasty rate, individual complication rates, ICU and hospital length of stay, return to OR, and readmission within 60 days.

Statistical Analysis:

Assumptions were checked before test selection. Categorical outcomes were compared with Fisher's exact test because several expected cell counts were below 5, and phi (φ) was reported as the effect size. Normality was assessed with the Shapiro-Wilk test and variance equality with the Brown-Forsythe test. Normally distributed continuous variables were compared with Student's t-test (Cohen's d); non-normal variables were compared with the Mann-Whitney U test (rank-biserial r). Analyses were performed in JASP (v0.98.8); α = .05.



RESULTS

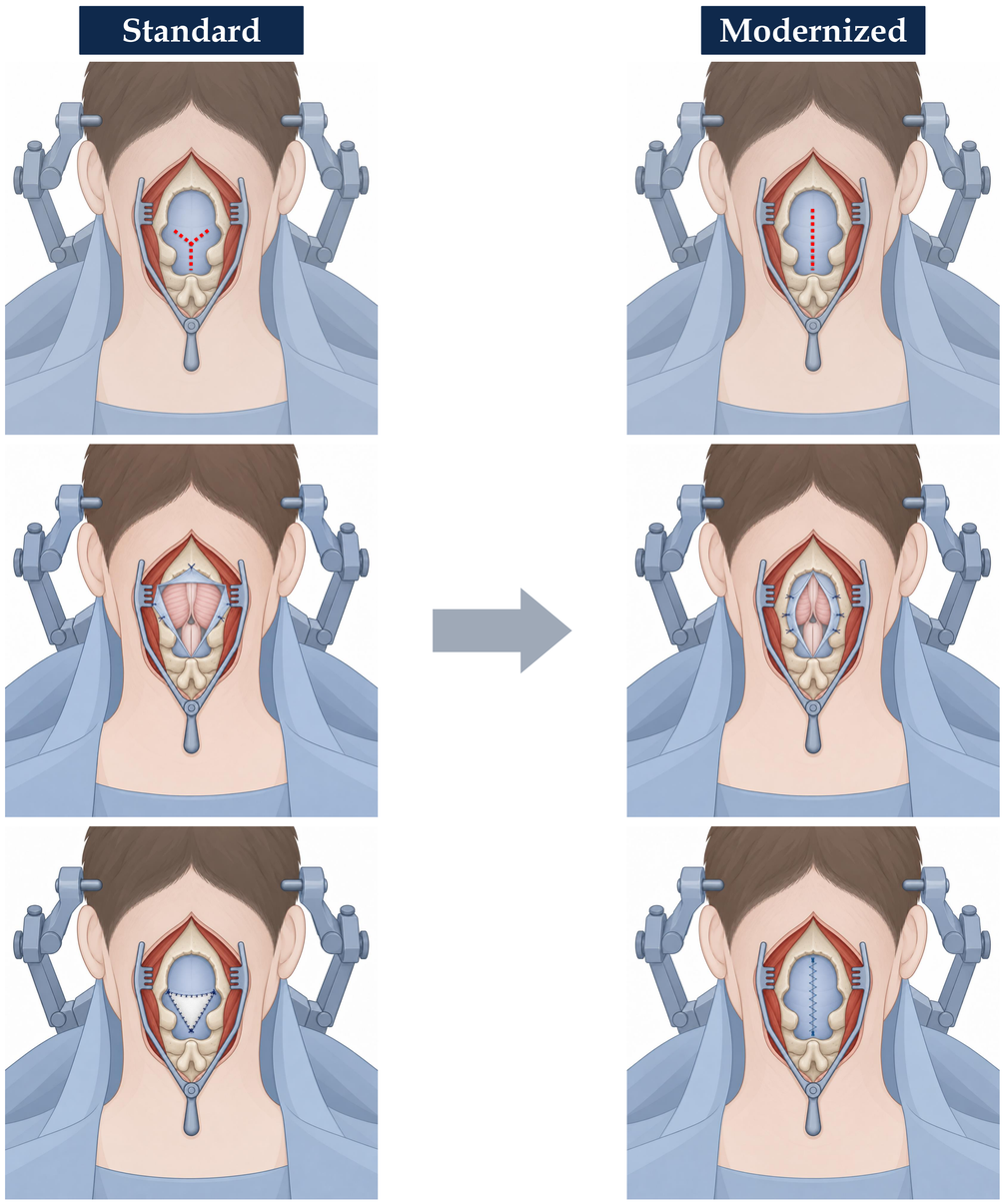


Figure 1. Standard Y-shaped dural incision, tonsillar exposure, and closure with duraplasty graft.

Figure 2. Modernized vertical linear dural incision, tonsillar exposure, and watertight primary closure.

KEY TAKEAWAYS

33.3%	**60-day complications** *p* = .063	**4.8%**
93.3%	**Duraplasty** ***p* = .024**	**57.1%**
20.0%	**CSF leak** *p* = .287	**4.8%**
2.69	**ICU stay (days)** *p* = .052	**1.26**

LEGEND

- **CM-I:** Chiari malformation type I
- **PFD:** posterior fossa decompression
- **Y-incision:** Y-shaped dural opening with three closure apices
- **Linear incision:** single vertical midline dural opening
- **Duraplasty:** graft incorporated into the dural closure to expand the subarachnoid space
- **LOS:** length of stay; **ICU:** intensive care unit
- **φ:** phi coefficient; **r_{rb}:** rank-biserial correlation

RESULTS

Variable	Y-incision (n = 15)	Linear (n = 21)	*p*
Age, yr	42.3 ± 10.4	42.4 ± 13.0	.991
Female, n (%)	13 (86.7)	18 (85.7)	> .999
Tonsillar descent, mm	13.3 ± 5.1	11.0 ± 4.1	.156
Symptom duration, d	329 ± 235	318 ± 333	.567
Syrinx, n (%)	6 (40.0)	8 (38.1)	> .999
Arachnoid opening, n (%)	13 (86.7)	21 (100)	.167
Tonsillar cautery, n (%)	11 (73.3)	21 (100)	.023
Duraplasty, n (%)	14 (93.3)	12 (57.1)	.024

Table 1. Baseline and operative characteristics. Values are mean ± SD or n (%). *p* values from Student's t-test (age, tonsillar descent), Mann-Whitney U (symptom duration; n = 12 vs 18), or Fisher's exact test.

RESULTS

- 60-day composite complications were lower in the linear group (1/21, 4.8%) than the Y group (5/15, 33.3%); *p* = .063.
- CSF leak was lower in the linear group (1/21, 4.8% vs 3/15, 20.0%; *p* = .287); pseudomeningocele occurred only in the Y group (3/15, 20.0%).
- All 6 complications occurred in patients who received duraplasty (6/26, 23.1%) vs 0/10 without duraplasty.
- ICU stay was shorter in the linear group (1.26 ± 0.45 vs 2.69 ± 2.43 d; *p* = .052).
- Hospital LOS did not differ (2.21 ± 1.23 vs 4.08 ± 4.39 d; *p* = .500).
- Return to OR and 60-day readmission each occurred in 1/21 linear vs 5/15 Y patients.

SUMMARY STATEMENTS

- The linear incision was associated with a 28.6-percentage-point lower 60-day composite complication rate (4.8% vs 33.3%), a difference that did not reach significance in this sample.
- Every complication occurred in a patient who received duraplasty, and the linear incision more often allowed primary closure without a graft. Reduced reliance on duraplasty may therefore mediate the lower complication burden, though this pathway was not directly tested.
- The shorter ICU stay in the linear group trended toward significance and is consistent with a less complicated early postoperative course.
- The lack of significance likely reflects limited statistical power (N = 36).
- These findings support further study of the linear incision for adult CM-I decompression.

REFERENCES

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